

three-dimensional environmental maps. Vision systems identify landmarks that help correct drift.

When GNSS signals are blocked or affected by multipath, sensor fusion helps maintain positioning continuity and filter errors, often preserving centimetre-level accuracy. Modern autonomous platforms, such as self-driving cars, warehouse robots, drones, and survey systems, do not rely solely on GNSS. Sensor fusion ensures continuous and reliable positioning in complex environments.

Reliability is as important as accuracy. High-precision GNSS performance depends on multiple system elements, including the operating environment, mechanical design, correction source, and overall system architecture.

In dense urban environments, tall buildings create severe multipath effects. Engineers address this by using multi-band, multi-constellation antennas (L1/L2/L5), integrating GNSS with IMUs or vision systems, and applying advanced filtering techniques such as Kalman filters. Real-time RTK or PPP processing also requires sufficient computing power.

In forested environments, limited satellite visibility weakens signals.



High-gain antennas, stable correction services, and sensor fusion help maintain reliable positioning.

High-motion platforms such as drones require faster update rates (typically 10-20Hz or higher), motion-tuned firmware, multi-constellation tracking, and intelligent filtering to maintain stable positioning.

Correction sources must also match the operating environment.

Local base stations provide low-latency RTK corrections. Cellular networks can support RTK or PPP when coverage is available. Satellite L-Band services provide wide-area corrections at a higher cost and with greater latency.

Finally, field testing remains essential. Open-sky performance alone is not sufficient. Systems must be validated in cities, forests, and industrial environments.

Achieving reliable centimetre-level accuracy requires a comprehensive approach that combines robust hardware, multi-band antennas, sensor fusion, dependable correction services, and thorough real-world validation.

Correction data: From theory to deployment

Centimetre-level GNSS performance depends on both the correction method and the operating environment.

Real-time kinematic (RTK) positioning remains the standard method for achieving fast ambiguity resolution and high repeatability within a defined region. RTK systems use a nearby base station or a continuously operating reference station (CORS) network to transmit real-time carrier-phase corrections. Under ideal conditions, RTK achieves horizontal accuracy of approximately one to two centimetres.

One common application is machine control in heavy construction. Contractors using systems from Trimble Inc. or Topcon Corporation integrate RTK positioning into dozers and graders.

On highway construction projects, centimetre-level control reduces both overcutting and under-cutting, saving significant material and labour costs. Even a two- to

Engineering checklist for next-generation positioning systems

When developing advanced positioning solutions, follow these principles:

- **Define accuracy requirements early.** Decide whether the application needs metre-, decimetre-, or centimetre-level precision. Let this requirement guide architecture, hardware, correction services, and budget.
- **Plan for GNSS limitations.** Expect reduced performance in urban canyons, dense foliage, near-indoor spaces, and high-dynamic conditions. Multipath, obstruction, and atmospheric effects will impact accuracy.
- **Choose the right architecture.** Select DGPS, RTK, PPP, or hybrid sensor fusion based on latency, correction access, connectivity, and environmental conditions.
- **Prioritise antenna and sensor integration.** Use high-quality multi-band, multi-constellation antennas. Combine GNSS with IMUs, lidar, vision, or wheel odometry through robust fusion methods.
- **Design for real-world conditions.** Address RF interference, vibration, temperature changes, signal blockage, connectivity loss, and environmental exposure from the start.
- **Validate in the field.** Test in real operating environments to confirm that lab performance holds up in practice.